

SatQuery AI

What We Did — Complete Technical & Feature Report

ISRO Problem Statement SIH26167 (Smart India Hackathon 2026) — Agentic Remote-Sensing Vision-Language Assistant. This report covers every model, service, feature, and honesty fix built across the project's full development history: 123 commits from 2026-08-24 to 2026-09-09.

1. Executive Summary

SatQuery AI lets an analyst ask natural-language questions over satellite imagery and get back a real, auditable answer: text, grounded bounding boxes, a pixel-level change mask, or a fused optical+SAR read — routed automatically to the right specialist model, with a confidence score and a downloadable evidence report. It is built around one core, genuinely fine-tuned vision-language model (Section 2, Model 1), four supporting specialist models (Sections 2.2–2.5), and a tiered fallback chain so the system degrades honestly instead of failing when no GPU is available (Section 2.6).

A large share of the project's effort — and a defining value of this team's process — went into repeated honesty audits: finding and removing fabricated data, hardcoded “demo” numbers, and overclaimed model attributions wherever they appeared, and replacing them with either a genuine computation or an honest “not available” state. Section 5 walks through these audit rounds. The newest feature, a judge-facing “Guide Bot” Q&A assistant, is covered in Section 7.

2. The Core AI Pipeline — Model by Model

The Problem Statement mandates at least one genuinely remote-sensing-adapted vision/vision-language model (a generic, non-adapted LLM/VLM alone does not satisfy it), single-image VQA, text-guided grounding, bi-temporal change analysis, and optical+SAR cross-modal fusion, all under one agentic orchestrator. Five specialist models plus a fallback chain implement this.

Model 1 — Qwen2.5-VL-3B + LoRA (the mandatory RS-adapted model)

File: backend/models/geochat_service.py. A 4-bit (NF4) quantized Qwen2.5-VL-3B with a custom LoRA adapter fine-tuned on VRSBench remote-sensing image/question/answer data — this is the model that satisfies the PS's mandatory adaptation requirement. It answers single-image VQA and, via its fine-tuned coordinate head, referring-expression grounding (“the water body in this image”). **Verified: 77.0% LLM-judged accuracy on a 200-question held-out VRSBench set.** Training took multiple real, documented debugging passes on free-tier Kaggle T4 GPUs — loss-masking bugs, a vision-tower/LoRA name collision, EOS/undertraining fixes, and a full second fine-tuning stage on real Sentinel-2 imagery.

Model 2 — Grounding-DINO + SAHI (open-vocabulary, multi-instance detection)

File: backend/models/grounding_dino_service.py. Added because VRSBench's own published paper confirms the fine-tuned Model 1 architecturally emits only one box per query — this model handles category-wide queries (“locate all buildings”) using an open-vocabulary detector with SAHI tiling for small/dense objects. **Verified: Acc@0.5 44.0%, Acc@0.7 16.0%, mean IoU 0.3671** — openly disclosed in the project's own research paper as still being actively hardened, not hidden.

Model 3 — AdaptFormer-CD / ChangeFormer (bi-temporal change detection)

File: backend/models/changeformer_service.py. A Siamese ViT with a lightweight adapter, evaluated against the published LEVIR-CD/WHU-CD benchmarks (**external baseline F1 91.4%** — the original authors' own published number, explicitly labeled as such, never claimed as a SatQuery-internal result). Produces a genuine per-pixel change-probability mask, polygonized via rasterio.features.shapes into real vector GeoJSON rendered on the map — replacing an earlier fixed 8×8 grid approximation. Real WGS84 polygon-area math converts this into hectares/acres for the SITREP report.

Model 4 — RemoteCLIP (zero-shot tagging & retrieval)

File: backend/models/remoteclip_service.py. Real zero-shot image/text embedding and tagging against a curated remote-sensing label taxonomy — powers catalog search, the Grid Anomaly Sweep, and the Confidence Stress-Test (Section 6).

Model 5 — SAR Fusion Module (optical + SAR cross-modal analysis)

File: backend/models/sar_fusion_service.py. Computes genuine NumPy-derived backscatter statistics directly from Sentinel-1 VV/VH pixel data — mean backscatter, standard deviation, water-likelihood ratio, urban double-bounce ratio, volume scattering ratio — fused with the vision-language model's reasoning across both images. This satisfies the PS's mandatory cross-modal pair-analysis requirement.

Fallback chain & router (used only when Model 1 isn't reachable)

Every specialist model above requires GPU access. When none is available, geochat_service.py tries, in order: (1) a local/Colab-tunnel GPU, (2) **Gemini** — a real vision-capable API call, honestly labeled “Generic Vision-Language Model (fallback tier — not remote-sensing fine-tuned)” and never allowed to set the is_real_model_loaded/adapted_active flags the PS's adaptation claim depends on, (3) **Groq**-hosted Llama-3 for text-only reasoning, (4) a deterministic heuristic as the last resort. router.py (Groq-assisted, with a rule-based fallback) classifies task/intent; validation.py checks image count, modality tags, and format before dispatch — together these meet the PS's mandatory agentic-orchestration requirement.

#	Model / Tier	Role	Status
1	Qwen2.5-VL-3B + LoRA (VRSBench)	Mandatory RS-adapted VQA + referring-expression grounding	Real, verified — 77.0% accuracy
2	Grounding-DINO + SAHI	Open-vocabulary, multi-instance detection	Real, verified — mIoU 0.367
3	AdaptFormer-CD / ChangeFormer	Bi-temporal change detection (CDVQA)	Real, verified — F1 91.4% (published baseline)
4	RemoteCLIP	Zero-shot tagging & retrieval	Real, verified
5	SAR Fusion Module	Optical + SAR cross-modal fusion	Real, verified
—	Gemini	Vision fallback tier (no GPU available)	Real, honestly labeled non-RS-adapted
—	Groq Llama-3	Text-only fallback + task router	Real, honestly labeled

Status colors: green = fully real and independently verified; amber = real but an intentionally limited fallback tier, always disclosed as such.

3. Supporting Backend Infrastructure

- **RAG grounding layer** (backend/rag/vector_store.py) — an in-house, in-memory NumPy cosine-similarity search over a curated knowledge base (not a dedicated vector database — an earlier claim naming “Qdrant Vector DB” was found and corrected during audit, see Section 5).

- **Live satellite catalog search** (`backend/geo/bhoonidhi_client.py`) — a real, unauthenticated Copernicus OData integration for live Sentinel-1/2 scene search, fixed from a fully hardcoded scene list. ISRO Bhoonidhi itself is shown as an honestly-labeled static example, since no public unauthenticated Bhoonidhi API exists.
- **Georeferencing** (`backend/models/geo_transform.py`) — never fabricates a coordinate transform; returns None unless the raster genuinely carries CRS metadata. Extended with a same-basename GeoTIFF sidecar mechanism scoped only to the project's own demo AOIs, so the bi-temporal hectare/acreage feature (Section 6) actually has real geometry to compute against.
- **SITREP audit report** (`backend/report_generator.py`) — downloadable PDF/JSON per query: sensor info, VQA response, bounding-box table, router decision trace, a real bi-temporal area-analysis section, and a SHA-256 content-integrity hash over the report's own recorded fields.
- **Conversational memory** (`backend/session_store.py`) — real, text-only multi-turn session history (question + answer + task), capped at 5 turns, threaded into every inference tier's prompt; images from prior turns are never re-attached.
- **Human-in-the-loop feedback** — a real analyst-correction queue (`/api/feedback`) persisted for future retraining, wired end-to-end to the HITL annotation page.
- **Hindi/English translation** — a domain-specific remote-sensing lexicon, wired to the language toggle and to voice input/output via the Web Speech API.

4. Frontend & UX Feature Set

React + Vite + Tailwind + React-Leaflet (2D only, by design, for low-bandwidth/mobile use). Confirmed genuinely wired to a real backend or external API:

- Natural-language query box with voice input (Web Speech API), sample-AOI selector, GeoTIFF/PNG upload, modality pills, bilingual Hindi/English toggle.
- Interactive Leaflet map with real bounding-box rendering and real ChangeFormer GeoJSON overlay.
- Side-by-side sensor comparison with a drag-to-compare swipe slider (optical/SAR or bi-temporal T1/T2).
- Real answer text, confidence %, computed bounding-box area, text-to-speech playback.
- Five “Mission Cockpit” domain profiles (ISRO, NDMA disaster, MNCFC/ICAR agriculture, MoHUA urban, INCOIS maritime), each pre-wiring a relevant AOI, modality, and default query.
- Real GeoJSON / KML / WKT / CSV export; live Copernicus catalog search modal.
- A real interactive API tester (Developer Hub) and a genuine live Situational Twin page (Open-Meteo weather, USGS earthquake feed, OpenStreetMap Nominatim geocoding).

Explicitly flagged as UI-only mockups (disclosed rather than presented as working features): the “XAI Vision Token Attention Heatmap” page (no backend calls; a hardcoded illustrative attention-percentage narrative) and the ministry/citizen-alert dashboard mockup inside Continuous Monitoring (the underlying “Top Anomalies” table itself is real and backend-wired; the ministry/SMS-alert portion is illustrative, no scheduler/notification backend exists yet).

5. The Honesty Audits — Finding & Fixing Fabrication

This team holds itself to a standard it calls, internally, “PS compromise nahi” — never fabricate a claim, a number, or a model attribution, and always distinguish a genuine result from an honest fallback. That standard was enforced through several dedicated audit rounds across the project's history, each one finding and fixing real issues rather than leaving them

in place for a demo:

Round	What was found & fixed
Early rounds (Aug 24–26)	Removed a fabricated pre-seeded response cache that was bypassing the real pipeline; fixed the root cause of an inaccurate “locate buildings” demo answer (the model never actually saw the image); fixed a router bug always reporting modality=Optical regardless of input; added a real Gemini vision fallback tier; fixed a coordinate-order bug in the grounding box parser.
Round (Aug 26–29)	Fixed fabricated eval/benchmark data and their generator scripts; fixed a CDVQA/Fusion compliance gap where the local RS-adapted model wasn't tried before the cloud VLM; added real SAHI+WBF tiled grounding; fixed real CRS/co-registration validation and cache-key correctness bugs found across two teammate audit rounds; replaced a fixed 8×8 grid change-mask approximation with real per-pixel polygonization.
Round (Sep 3–4)	Fixed fabricated benchmark claims across the UI; corrected a stale VQA accuracy figure (84.0%) that contradicted the verified 77.0% number used everywhere else; corrected a “Qdrant Vector DB” architecture claim to match what the code actually runs (an in-house NumPy vector store).
This session (Sep 9)	Independently verified a teammate's claim of “7 successfully implemented and live-tested features” and found 4 of the 7 (Investigation Orchestrator, Grid Sweep, Time-Machine, Stress-Test) were fully fabricated — including specific fake legal-violation citations naming real government agencies. Rewrote all 4 to call the real models genuinely, with honest “SKIPPED / insufficient evidence” states when real imagery isn't available. Also found and fixed a fake “Attention Heatmap” overlay in the map view, and a fake standby telemetry screen in the Audit Trace modal shown before any query ran.

A further review of the team's own Vercel/Gemini demo integration (also this session) found the hosted demo's serverless gateway currently returns keyword-templated bounding boxes and a fixed model-attribution list rather than deriving them from that specific image — documented plainly, with the fix scoped for a future round, in docs/GUIDE_BOT_KNOWLEDGE.md (Section 7) so the system itself discloses this rather than overstating what the live demo does.

6. Deployment Architecture

Two ways to run SatQuery, and it matters which one a viewer is looking at:

A) Full stack (the real pipeline, Models 1–5)

FastAPI backend on a GPU host — local machine, Render, or a Colab/Kaggle-tunneled notebook — with the frontend's VITE_API_URL pointed at it. This is what genuinely exercises the fine-tuned Qwen2.5-VL LoRA, ChangeFormer, RemoteCLIP, and SAR fusion pipeline described in Section 2.

B) Hosted Vercel demo (frontend + one serverless function)

The public demo link runs only the React frontend plus api/vlm.js, a serverless function that calls Gemini directly through a multi-key failover pool. The Python/PyTorch backend is **not** deployed here — every query on this specific link is answered by Gemini, not by Models 1–5. This is disclosed plainly (rather than left implicit) in both this report and the Guide Bot's own knowledge base, per this team's honesty standard.

Confidence Stress-Test (backend/stress_test.py) is a related, genuinely real capability worth calling out on its own: it perturbs the actual query image (rotation, crop/zoom, brightness/contrast jitter) and re-runs real RemoteCLIP tagging on each variant to measure label stability — a real robustness check, not a canned number, whenever it has a real image to work with.

7. Newest Feature — The SatQuery Guide Bot

A judge-facing Q&A assistant, added this session, so anyone evaluating the project can ask about any feature or the technical architecture and get an answer grounded in the team's own documentation — not improvised. Three parts:

- docs/GUIDE_BOT_KNOWLEDGE.md — a comprehensive knowledge base covering the PS requirements, the real pipeline (this report's Section 2), what's actually live in the Vercel demo vs. what needs the full backend (Section 6), frontend real-vs-mockup status, and explicit answering rules (never invent a fact; state “not verified” rather than guess).
- api/guide.js — a Vercel serverless endpoint using the same multi-key Gemini failover pattern as the main VLM gateway. Unlike that gateway, this endpoint **never fabricates an answer**: any failure returns a plain “couldn't reach the assistant” message instead of a plausible-sounding invented one — the one feature whose entire job is describing the system honestly cannot itself repeat the mistake documented in Section 5.
- GuideBotModal.jsx — a chat UI with starter prompts, multi-turn history, and a visible “powered by Gemini, grounded on our own docs” disclaimer. Reachable from a header button and an always-visible floating launcher on every tab.

8. Known Limitations (Disclosed, Not Hidden)

- The public Vercel demo's bounding boxes are currently keyword-templated, not derived from that specific image — the model-attribution list it returns names specialist models that aren't actually called on that code path. Scoped for a future fix; documented in the Guide Bot's own knowledge base today.
- AR Ground-Truth Viewfinder and Satellite Time-Machine's illustrative epoch panel currently show representative, not per-scene-measured, overlay content when real per-frame data isn't available — a UX concept demo, not yet wired to live geospatial computation for that exact camera frame.
- Grounding accuracy (Acc@0.7 16.0%) is real but still being actively hardened — disclosed as such rather than rounded up.
- Two frontend pages (XAI Attention Heatmap, the ministry/citizen alert dashboard mockup) are UI-only and should be demoed as concepts, not as live AI capabilities, until wired to a real backend.
- NDVI/NDWI spectral-index toggle is blocked on sourcing real multi-band Sentinel-2 imagery — the demo AOIs are RGB-only, and approximating NDVI from RGB would itself be a fabricated-capability claim.

9. Verified Benchmark Numbers (Reference)

Capability	Metric	Result	Source
VQA (Model 1)	LLM-judged accuracy	77.0%	200 held-out VRSBench questions
Grounding (Models 1+2)	Acc@0.5	44.0%	VRSBench held-out eval
Grounding (Models 1+2)	Acc@0.7	16.0%	VRSBench held-out eval (disclosed as still hardening)
Grounding (Models 1+2)	Mean IoU	0.3671	VRSBench held-out eval, X-first coordinate order
Change Detection (Model 3)	F1 score	91.4%	Published LEVIR-CD/WHU-CD baseline (external, not internal)
Bi-temporal area calc	Verified example	852.155 ha	Brahmaputra flood AOI, cross-checked against 13.68% change rate

10. Project Timeline & Scale

Metric	Value
Development window	2026-08-24 → 2026-09-09 (17 days)

Metric	Value
Total commits	123
Core specialist models	5 (plus a 3-tier honest fallback chain)
Backend services	14+ (models, RAG, geo, translation, session store, report generator, cache)
Frontend pages/components	35+
Dedicated honesty-audit rounds	6+ (Aug 24, Aug 26, Aug 29, Sep 3–4, Sep 9 ×2)
Public API endpoints	16 (FastAPI) + 2 (Vercel serverless: VLM gateway, Guide Bot)

This report was compiled directly from the project's own git history and its own audit-verified feature documentation. Where a capability is a fallback, a demo illustration, or still being hardened, it is labeled as such above rather than presented as equivalent to a fully verified result.